Real-Guided Skeleton Augmentation for Actuator-Constrained Rock-Paper-Scissors Robot Counterattack

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***Abstract*—This report presents a skeleton-first real-to-sim-to-real pipeline for an embodied Rock-Paper-Scissors counterattack task. The project begins with a small set of real hand-motion videos, extracts MediaPipe 21-landmark skeletons, canonicalizes the trajectories, and expands them into real-guided synthetic skeleton datasets aligned to SCHUNK/Isaac-style robot-hand pose progress. From only 20 reviewed real clips and 720 processed frames, the pipeline builds a 10,000-sample sharded skeleton dataset and a render-ready 32-frame alignment layer. The final live validation applies the frozen v4 temporal predictor back to a prompt-window camera capture and produces an actuator-feasible robot counterattack with 0.4667 s remaining. The robot hand is therefore the final embodied response target, while the main research contribution is the real-guided skeleton augmentation and transfer path.**

***Index Terms*—real-to-sim, sim-to-real, skeleton sequence modeling, MediaPipe, early action prediction, actuator feasibility, robot hand**

I. Introduction

Embodied interaction requires more than recognizing a final pose after an action is complete. In a Rock-Paper-Scissors counterattack, a robot must infer the human's intended gesture early enough to choose a winning response and still move its hand before the actuator deadline. This makes the problem a combined data, prediction, and control problem: sparse real hand-motion data must support an early temporal predictor, and the predictor must transfer back to live camera input under timing constraints.

The main difficulty in this project is data scarcity. Recording many high-quality real hand-motion videos is expensive, and final-frame pose labels do not capture when the hand begins to reveal the final gesture. The project therefore uses a skeleton-first real-to-sim-to-real strategy. A small set of real videos is converted into MediaPipe 21-landmark trajectories, those trajectories are canonicalized into hand-local coordinates and temporal features, and real-guided augmentation generates broader skeleton motion coverage. The synthetic skeletons are aligned to SCHUNK/Isaac-style pose progress so they remain connected to the intended robot-hand embodiment.

Fig. 1 summarizes the final pipeline. The pipeline is deliberately not a raw image-to-landmark Isaac perception system. Instead, it follows the project boundary of skeleton ground truth first: camera frames are used to obtain MediaPipe skeletons,

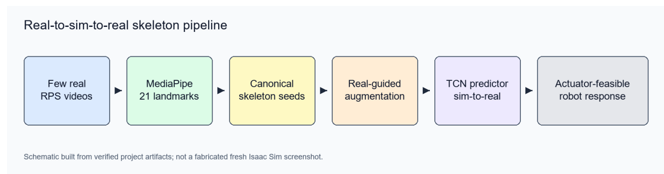

Fig. 1. Real-to-sim-to-real pipeline schematic. The figure is a schematic built from verified project artifacts, not a fresh Isaac Sim screenshot.

and the temporal model consumes canonical skeleton features. The robot hand appears at the end as an actuator-constrained response target.

This report makes four contributions:

- a few-shot real RPS skeleton seed dataset extracted from MediaPipe 21-landmark hand tracks;
- a real-guided skeleton augmentation pipeline with compact and large sharded datasets;
- a render-ready SCHUNK/Isaac alignment layer that improves pose-progress quantization over the earlier 8-frame grid;
- a sim-to-real prompt-window counterattack demo using the frozen v4 predictor and actuator feasibility checks.

II. Related Work

Early action prediction differs from ordinary action recognition because the model must decide before the full motion is observed [1], [2]. This project narrows that setting to prompt-conditioned hand motion, where useful evidence is distributed across the transition from a starting fist to the final RPS pose. The final-frame class is not sufficient: the decision must arrive early enough for a robot hand to respond.

Skeleton-based temporal models are a practical fit for this setting. GRU and TCN-style models can process short sequential feature vectors without requiring a full raw-video perception stack [3]. This is why the project uses MediaPipe to extract 21 landmarks and then trains on canonical skeleton coordinates, velocities, finger-curl values, and motion signals. Selective prediction and calibration ideas also matter, because the system should not issue a counterattack when confidence, margin, or confirmation gates are weak [4], [5].

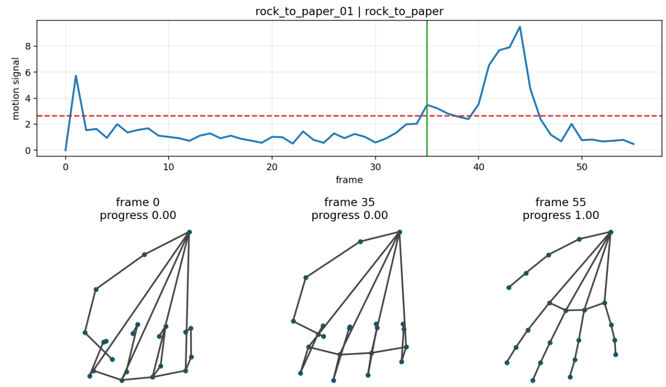

Fig. 2. Canonical MediaPipe skeleton seed example. Real videos are converted into hand-local skeleton trajectories with motion signal, onset, and progress metadata.

Isaac Sim and Isaac Lab provide the target context for articulated robot assets and actuator modeling [6], [7]. In this submission, Isaac/SCHUNK work is used as render and robot-response grounding. The final learning path remains skeleton-first, while the SCHUNK-style hand is used to represent the robot response and to check whether a selected counter gesture is actuator feasible.

III. Method

A. Real MediaPipe Skeleton Seeds

The first stage records a small real-video seed set and extracts MediaPipe Hands landmarks. The verified seed review contains 20 real clips, split evenly between `rock_to_paper` and `rock_to_scissors`. MediaPipe returned 21 landmarks for every processed frame, giving 720 processed frames and minimum detection coverage of 1.0. Multiple-hand detections were preserved in metadata, but the highest-confidence hand was used as the primary hand.

B. Canonical Skeleton Features

The extracted landmarks are converted into canonical hand-local coordinates. The wrist is the origin, wrist-to-middle-MCP distance defines the primary scale, and the palm axes define a local coordinate frame. Each frame stores canonical 21-point coordinates, velocities, five finger-curl values, five tip-to-MCP distances, five fingertip speeds, and one motion signal. The resulting feature vector has dimension 142. Fig. 2 shows a representative canonical seed preview with motion signal and progress checkpoints.

C. Real-to-Sim Skeleton Augmentation

The canonical real skeleton seeds drive the synthetic expansion stage. A compact bulk dataset generates 2,000 augmented samples, balanced between `rock_to_paper` and `rock_to_scissors`. A larger sharded dataset generates 10,000 samples with a 7,000/1,500/1,500 train/validation/test split. The augmentation is intentionally real-guided: temporal

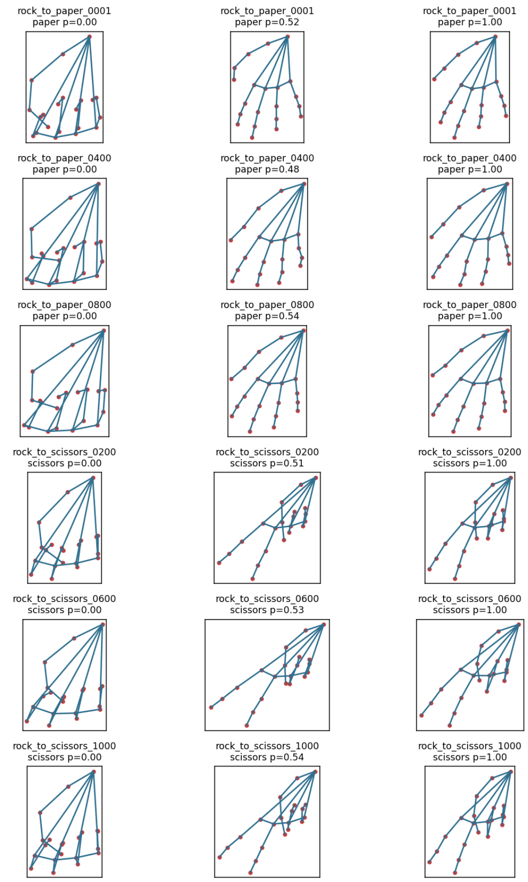

Fig. 3. Real-guided skeleton augmentation evidence. Few real MediaPipe skeleton seeds are expanded into multiple canonical skeleton trajectories for `rock_to_paper` and `rock_to_scissors`.

scaling, temporal warping, finger-specific scaling, small rotations, and landmark noise are applied to canonical skeleton trajectories rather than to arbitrary hand drawings.

D. SCHUNK/Isaac Render-Ready Alignment

The augmented skeleton sequences are aligned to a SCHUNK/Isaac-style robot-hand pose schedule. The older connected-motion evidence used an 8-frame progress grid; the real-guided render-ready alignment prepares a 32-frame schedule with 64 future render-manifest entries, 32 for each transition. This stage does not claim that every sample has a fresh rendered image. Instead, it stores per-frame lookup arrays that make the skeleton dataset ready to consume future SCHUNK render records.

Fig. 4 shows the alignment improvement. The maximum motion-progress quantization error drops from about 0.106 with the old 8-frame grid to about 0.016 with the 32-frame real-guided schedule.

E. Sim-to-Real Counterattack

After training and validation iterations, the final live/demo predictor is frozen to v4. It uses the rock-guard realtime selector config, profile weights $[0.25, 0.75, 0.0]$, confidence

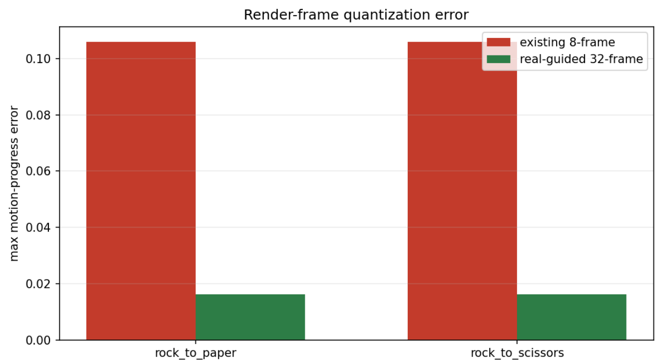

Fig. 4. Render-ready alignment improvement. The 32-frame real-guided schedule reduces the maximum motion-progress quantization error relative to the earlier 8-frame SCHUNK progress grid.

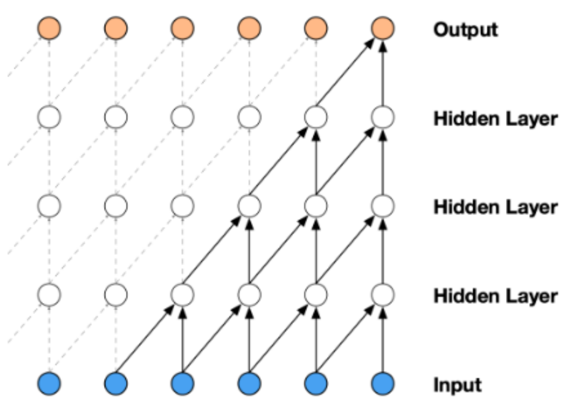

Fig. 5. Temporal predictor schematic. The model consumes a sequence of canonical skeleton features and predicts the response-window gesture before the robot-hand deadline.

threshold 0.70, margin threshold 0.10, confirmation count 2, and response prompt `scissors`. v7e is retained as diagnostics only and is not promoted. Fig. 5 shows the temporal convolution style of the sequence predictor as a conceptual model diagram, not as experimental evidence by itself.

The live prompt sequence is:

$$\texttt{rock} \rightarrow \texttt{paper} \rightarrow \texttt{scissors}.$$

The human performs the target gesture during the final `scissors` response window. The counter policy maps human rock to robot paper, human paper to robot scissors, and human scissors to robot rock. The robot starts from rock, waits for a confirmed response-window prediction, and then checks response delay, deadline, velocity, and normalized joint limits before accepting the episode as actuator feasible.

F. Actuator Timing Model

The counterattack is evaluated in the response window, not over the entire video. Let t_0 be the start time of the final response prompt, t_d be the first confirmed decision time, and

TABLE I

REAL-GUIDED SKELETON DATASET PIPELINE

Stage	Size	Evidence
MediaPipe seed review	20 clips, 720 frames	min detection coverage 1.0
Canonical seed dataset	20 sequences	max length 56, feature dim 142
Compact augmentation	2,000 samples	1,000 per transition
Large sharded dataset	10,000 samples	split 7,000 / 1,500 / 1,500
Render-ready alignment	64 manifest entries	32 frames per transition

TABLE II

SCHUNK/ISAAC RENDER-READY ALIGNMENT

Alignment grid	Max progress error	Role
Existing 8-frame grid	about 0.106	coarse connected-motion reference
Real-guided 32-frame grid	about 0.016	render-ready alignment layer

D be the actuator deadline after the response window opens. The response-window decision latency is

$$L_d = t_d - t_0,$$

and the remaining actuator budget is

$$R = D - L_d.$$

The robot response is feasible only if the actuator result satisfies

$$\tau_{\text{req}} \leq R$$

and all normalized joint targets remain inside their configured limits. Here τ_{req} already includes the configured response delay and the slowest required joint motion. In the final selected live take, $t_0 = 2.0$ s, $t_d = 2.0333$ s, $L_d = 0.0333$ s, $D = 0.50$ s, and $R = 0.4667$ s. The configured response delay is 0.08 s, the total required robot response time is 0.4133 s, the limiting joint is `index_curl`, and the episode passes because $0.4133 \leq 0.4667$.

IV. EXPERIMENTS

Table I summarizes the real-to-sim dataset path. The key point is that the project does not jump directly from a few real videos to a final demo. It first constructs a verified skeleton representation, then expands it into larger real-guided synthetic datasets.

Table II summarizes the alignment improvement. This table is important because it connects skeleton augmentation to the later robot-hand embodiment. The large skeleton dataset remains usable for skeleton-only training, while the render-ready fields make the same samples compatible with future SCHUNK/Isaac render-conditioned experiments.

A. Evaluation Platform and Runtime Context

The live timing evidence should be interpreted as local end-to-end validation timing. It includes camera capture, MediaPipe hand tracking, canonical feature construction, PyTorch model inference, decision gating, OpenCV overlay writing, and the feasibility computation on the local desktop used for recording. It is not a hardware-independent model-latency

TABLE III

LOCAL RUNTIME ENVIRONMENT FOR FINAL LIVE VALIDATION

Component	Value
Operating system	Windows 11 Pro 10.0.26200
CPU	Intel Core i5-14400F, 10 cores / 16 logical processors
Memory	34164101120 bytes, about 31.8 GiB
GPU	NVIDIA GeForce RTX 4060, 8188 MiB, driver 560.94
Runtime policy	`device=auto`; CUDA is available for PyTorch model execution

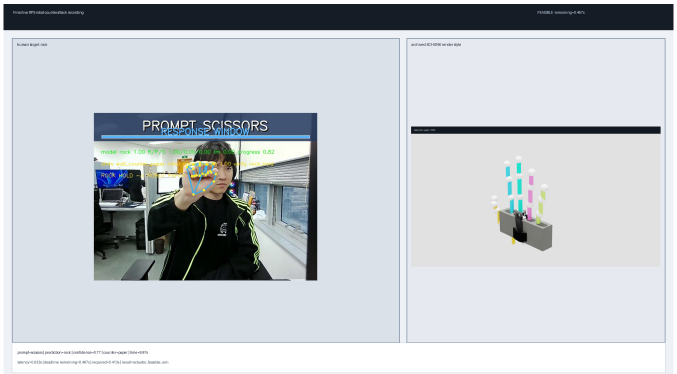

Fig. 6. Final sim-to-real counterattack validation. The frozen skeleton predictor is applied back to live prompt-window input and drives an actuator-feasible robot paper response.

benchmark. The realtime configuration uses `device=auto`; on this machine PyTorch can select CUDA because an NVIDIA GPU is available, while MediaPipe and OpenCV remain part of the measured loop.

V. FINAL LIVE VALIDATION

The final live validation tests whether the skeleton predictor transfers back to a real prompt-window camera capture. The selected final take is `human rock -> robot paper`. During the final `PROMPT SCISSORS` response window, the system predicts rock and issues robot paper. The actuator feasibility checker reports an actuator-feasible win with 0.4667 s remaining. Fig. 6 shows the poster frame.

Fig. 7 visualizes the selected take's timing budget. The decision occurs shortly after the response window opens, leaving enough time for the delayed robot command and the rock-to-paper motion.

The robot-hand visuals use archived SCHUNK-style render keyframes as the response pose library, shown in Fig. 8. These images explain the screen-rendered robot response, but they are not the main evidence for the real-guided data expansion.

Table IV records the final model and demo policy. The v4 fallback remains the live/demo predictor. v7e reached original20 strict validation at 17/20 and is retained only as diagnostic evidence; it is not promoted and v7f retraining is not started.

VI. LIMITATIONS

The most important result is not that a screen-rendered robot hand appears in the final video. The important research path

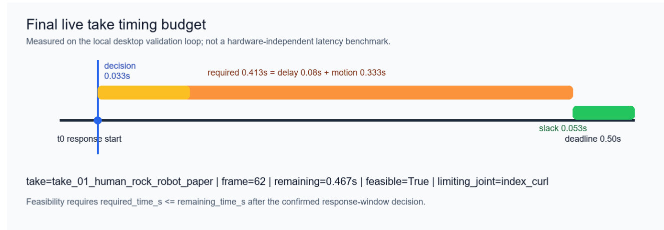

Fig. 7. Final selected take timing budget. The measured decision latency is local end-to-end validation timing on the desktop in Table III, not a universal model latency number.

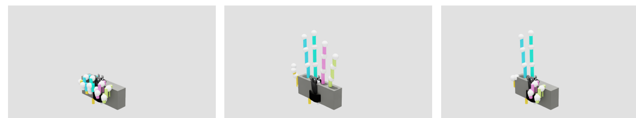

Fig. 8. Archived SCHUNK-style response pose library for rock, paper, and scissors. These verified keyframes support the screen-rendered robot response layer.

is that a small real skeleton seed set can be converted into a larger real-guided skeleton dataset, aligned to the robot-hand embodiment, and then transferred back to live camera input for a timing-constrained response.

The project also exposes several limitations. First, the final training and live/demo path is skeleton-first. The render-ready sidecars prepare the dataset for image- or render-conditioned Isaac experiments, but the final predictor does not learn from raw Isaac camera images. This is an intentional v1 boundary rather than a hidden failure. It keeps the final system verifiable and avoids confusing skeleton prediction with a separate vision-based landmark-estimation problem.

Second, the final timing values depend on the local validation platform. The measured latency includes camera acquisition, MediaPipe, feature extraction, PyTorch/CUDA inference when available, decision gating, and OpenCV video output. A different camera, CPU, GPU, driver stack, or display path could change the end-to-end timing even if the learned predictor weights are unchanged.

Third, only one of the final prompt-scissors live takes passed the full actuator-feasible counterattack criterion. The paper take failed because no confirmed paper decision was produced inside the response window. The scissors take produced a confirmed scissors decision, but too late for the actuator deadline. These failures are useful because they show why the evaluation must include timing and feasibility, but they also limit the breadth of the live evidence.

VII. FUTURE WORK

The first extension is hardware-normalized latency benchmarking. Future runs should separate MediaPipe time, feature construction time, model forward-pass time, decision-gate time, and rendering/output time, then report them across CPU-only and CUDA-enabled configurations. This would distinguish predictor speed from local camera and video I/O overhead.

TABLE IV
FINAL PREDICTOR AND LIVE DEMO METRICS

Category	Metric	Value
v4 fallback	profile weights	$[0.25, 0.75, 0.0]$
v4 fallback	original20 strict	$20/20$
v4 fallback	heldout15 validation	$11/15$, validation-only
v7e diagnostics	original20 strict	$17/20$, not promoted
Final live take	selected take	human rock -¿ robot paper
Final live take	episode result	actuator-feasible win
Final live take	remaining time	0.4667 s

TABLE V
FINAL PROMPT-SCISSORS LIVE TAKE DIAGNOSTICS

Take	Decision	Outcome
human rock -¿ robot paper	predicted rock at frame 62; latency 0.0333 s	selected; actuator-feasible win; remaining 0.4667 s
human paper -¿ robot scissors	no confirmed paper response-window decision	diagnostic failure; no counter issued
human scissors -¿ robot rock	predicted scissors at frame 81; latency 0.6667 s	diagnostic failure; infeasible due negative time budget

The second extension is image-conditioned Isaac or SCHUNK training. The current dataset already contains render-ready alignment fields, but the final model still consumes skeleton features. A future branch could train an image-to-skeleton or render-conditioned temporal model, using the existing skeleton labels as supervision while measuring the cost of moving beyond the skeleton-ground-truth-first assumption.

The third extension is a more robust prompt-window model branch. The diagnostic v7e branch is retained only as report evidence after failing original20 validation at $17/20$, and v7f retraining was not started in this submission. A future v7f-style branch should begin only with explicit approval and should target the observed live failures: paper response-window confirmation and scissors-boundary timing under the actuator deadline.

Finally, the screen-rendered robot response should be transferred to a physical or higher-fidelity simulated hand once the perception and timing metrics are stable. Physical deployment would require actuator calibration, command latency measurement, and safety limits beyond the normalized kinematic feasibility check used here.

VIII. CONCLUSION

This project implements a real-guided skeleton augmentation pipeline for actuator-constrained RPS counterattack. It starts from 20 real MediaPipe-reviewed clips, builds canonical skeleton seeds, expands them into compact and 10,000-sample real-guided datasets, aligns the data to SCHUNK/Isaac-style pose progress, and validates the frozen v4 predictor back on live prompt-window input. The final robot-hand counterattack is the application-level proof that the prediction and timing policy can drive an actuator-feasible response, while the dataset and alignment pipeline form the central research contribution.

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